

(Applications Development and Emerging Technologies)

TECHNICAL-SUMMATIVE ASSESSMENT

2

PHP ARRAYS AND PRE-DEFINED FUNCTIONS

Student Name / Group Name:		
Members (if Group):	Name	Role
Section:		
Professor:		

I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Understand and apply best practices and standards in the development of website.

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- To know the different approach of using arrays in PHP.
- To use PHP lazy function foreach to iterate through array elements.
- To implement one dimensional and multi-dimensional in the program.
- To know use variables that is globally declared and locally declared.
- To know how to include separate PHP code in the main page for code enhancement.
- To be familiar with the use of common predefined function such as define, include, and require.
- To use different available mathematical function for manipulating numbers.

IV. BACKGROUND INFORMATION

PHP Arrays and User Defined Functions

Array: one dimensional

Example: Code 1

```
<?php
    $fruit[0] = "orange";
    $fruit[1] = "apple";
    $fruit[2] = "grapes";
    $fruit[3] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 2

```
<?php
    $fruit[] = "orange";
    $fruit[] = "apple";
    $fruit[] = "grapes";
    $fruit[] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 3

```
<?php
    $fruit = array("orange", "apple", "grapes", "banana");
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => orange
    [1] => apple
    [2] => grapes
    [3] => banana
)
```

PHP Predefined Functions

Using Constant

`define()` functions

used to declare constants. A constant can only be assigned a scalar value, like a string or a number. A constant's value cannot be changed.

Syntax:

```
define('NAME', 'value');
```

Example:

```
<?php
define('MAX_VALUE',10);
for($i=5;$i<MAX_VALUE;$i++){
    echo $i." ";
}
//MAX_VALUE = 20; is invalids
?>
```

Output:

5 6 7 8 9

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

1. **Using ARRAYS**, create a fruit directory that will show the image, name, description and facts about the fruit (create at least 10 fruits sort in alphabetical order). Implement a HTML and CSS code for the structure and design of the webpage. Please see the sample output below.

My Fruits			
Image	Name	Description	Facts
	Banana	Color Yellow	Bananas are a healthful addition to a balanced diet, as they provide a range of vital nutrients and are a good source of fiber.

2. **USING USER DEFINED FUNCTION.** Create a formula for each volume of shapes (Cube, Right Rectangular Prism, Prism or Cylinder, Pyramid or Cone and Sphere) then display the result. Please see the sample below:

Volume of Shapes		
Values	Formula	Answer
$s = 5$	$V=s^3$	125

3. Convert your Student Resume into a web based form, used include() and require() functions to connect all pages

Include and create the following pages:

- Career Objective page
- Personal information page
- Educational Attainment page
- Skills page
- Affiliation page
- Work Experience Page

Please see the sample below:

	Personal information
• Career Objective	
• Educational Attainment page	
• Skills page	
• Affiliation page	
• Work Experience Page	

VII. QUESTION AND ANSWER

1. What is Array?
2. What is function in PHP?
3. Differentiate the one-dimensional array to multi-dimensional array
4. Describe the difference between Global Variable to Local Variable
5. What is the difference between function without parameter and a function with parameter?
6. What are the different common predefined functions? Describe each and give example
7. Is it useful to use an include or require function on a webpage? Why?
8. What is the importance of a function?
9. What is the importance of a string function?

VIII. REFERENCES

1. <https://www.w3schools.com/css/>
2. <https://www.w3schools.com/html/>
3. https://www.w3schools.com/php/php_variables.asp
4. https://www.w3schools.com/php/php_arrays.asp
5. https://www.w3schools.com/php/php_arrays_indexed.asp
6. https://www.w3schools.com/php/php_arrays_associative.asp
7. https://www.w3schools.com/php/php_arrays_multidimensional.asp
8. https://www.w3schools.com/php/php_arrays_sort.asp
9. https://www.w3schools.com/php/php_functions.asp
10. https://www.w3schools.com/php/php_constants.asp
11. https://www.w3schools.com/php/php_includes.asp
12. https://www.w3schools.com/php/php_string.asp

Note: The following rubrics/metrics will be used to grade students' output.

Program (100 pts.)	(Excellent)	(Good)	(Fair)	(Poor)
Program execution (20pts)	Program executes correctly with no syntax or runtime errors (18-20pts)	Program executes with less than 3 errors (15-17pts)	Program executes with more than 3 errors (12-14pts)	Program does not execute (10-11pts)
Correct output (20pts)	Program displays correct output with no errors (18-20pts)	Output has minor errors (15-17pts)	Output has multiple errors (12-14pts)	Output is incorrect (10-11pts)
Design of output (10pts)	Program displays more than expected (10pts)	Program displays minimally expected output (8-9pts)	Program does not display the required output (6-7pts)	Output is poorly designed (5pts)
Design of logic (20pts)	Program is logically well designed (18-20pts)	Program has slight logic errors that do no significantly affect the results (15-17pts)	Program has significant logic errors (3-5pts)	Program is incorrect (10-11pts)
Standards (20pts)	Program code is stylistically well designed (18-20pts)	Few inappropriate design choices (i.e. poor variable names, improper indentation) (15-17pts)	Several inappropriate design choices (i.e. poor variable names, improper indentation) (12-14pts)	Program is poorly written (10-11pts)
Delivery (10pts)	The program was delivered on time. (10pts)	The program was delivered a day after the deadline. (8-9pts)	The program was delivered two days after the deadline. (6-7pts)	The program was delivered more than two days after the deadline. (5pts)